

## Dengue fever in the Western Hemisphere.

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Dengue virus, an arthropod-borne viral agent, causes two distinct diseases: classic dengue fever (DF) and dengue hemorrhagic fever (DHF). There are four dengue virus serotypes: DEN-1, DEN-2, DEN-3, and DEN-4. Although infection with dengue stimulates immunologic response to a serotype, there is no cross-immunity conferred. Hence, a person can potentially be infected with each serotype during his or her lifetime. An infected female *Aedes* mosquito transmits the virus from person to person while feeding. The disease, now endemic in more than 100 countries in Africa, the Americas, the Eastern Mediterranean, Southeast Asia, and Western Pacific, is spreading to new areas and causing explosive outbreaks. Because of the major impact on lives and local economies epidemics produce, rapid detection of dengue infection has become an important public health research issue. Recently developed serological procedures to detect dengue infections have shown great potential for field use.

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## Control of dengue virus in the midgut of *Aedes aegypti* by ectopic expression of the dsRNA-binding protein Loqs2.

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Dengue virus (DENV) is an arbovirus transmitted to humans by *Aedes* mosquitoes<sup>1</sup>. In the insect vector, the small interfering RNA (siRNA) pathway is an important antiviral mechanism against DENV<sup>2-5</sup>. However, it remains unclear when and where the siRNA pathway acts during the virus cycle. Here, we show that the siRNA pathway fails to efficiently silence DENV in the midgut of *Aedes aegypti* although it is essential to restrict systemic replication. Accumulation of DENV-derived siRNAs in the midgut reveals that impaired silencing results from a defect downstream of small RNA biogenesis. Notably, silencing triggered by endogenous and exogenous dsRNAs remained effective in the midgut where known components of the siRNA pathway, including the double-stranded RNA (dsRNA)-binding proteins Loquacious and r2d2, had normal expression levels. We identified an *Aedes*-specific paralogue of loquacious and r2d2, hereafter named loqs2, which is not expressed in the midgut. Loqs2 interacts with Loquacious and r2d2 and is required to control systemic replication of DENV and also Zika virus. Furthermore, ectopic expression of Loqs2 in the midgut of transgenic mosquitoes is sufficient to restrict DENV replication and dissemination. Together, our data reveal a mechanism of tissue-specific regulation of the mosquito siRNA pathway controlled by Loqs2.

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## Identification of dengue virus in *Aedes* mosquitoes and patients' sera from Si Sa Ket Province, Thailand.

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Dengue fever (DF), dengue hemorrhagic fever (DHF) and dengue shock syndrome (DSS) are the re-emerging infectious diseases caused by dengue (DEN) virus, transmitted by *Aedes* mosquito. There are more than 100,000 cases of dengue infection and more than 100 deaths annually in Thailand. Virological surveillance for DEN viruses is used as an early warning system to predict outbreaks. The seroprevalence of infection and serotypes of DEN virus in 116 pediatric patients at Si Sa Ket Province, Thailand were analyzed during June to September 2004. At the same period, *Aedes* mosquitoes were

caught from patients' and their neighbors' houses, from control houses, located in villages with no report of dengue infection during the previous 3 years. The majority of DHF cases were secondary infections of DEN-2 and DEN-4 serotypes. Of the 1,652 Aedes mosquitoes collected 1,583 were Ae. aegypti and 69 Ae. albopictus. Ten mosquitoes from each house were pooled and dengue viruses were determined using RT-PCR assay; only 1 positive pooled was found. Although the dengue infection rate in the field caught mosquitoes was low, the existing dengue virus control program in transmission areas by aerial spraying to destroy the larva breeding sites should be continued.

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## Assessment of multifunctional activity of graphene oxide and guar gum-based nanomaterials against Aedes aegypti.

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Nanomaterials have been receiving much research attention in controlling insect pests and vectors. Properties, such as small surface-to-volume ratio, low dosages, durability, solubility, enhanced target activity, pore size and surface characteristics have enabled the design of precise and targeted insecticides through adsorption, encapsulation, and conjugation. The reported study aims to evaluate the efficacy of graphene oxide (GO) and guar gum (GG)-based nanomaterials against early fourth instar of Aedes aegypti, a mosquito responsible for transmitting diseases like dengue fever, Zika, and chikungunya, among many others. GO and a total of eight nanomaterials (SA-1 to SA-8) were formulated using GO, GG and their combinations with ferrous oxide and copper oxide. Each material was characterized using various biophysical techniques. The materials were evaluated for larvicidal potential by assessing their efficacy on the survival and morphology of Ae. aegypti, while the contact irritancy activity focused on determining their irritant properties upon direct exposure to the female adults. The larvicidal and irritant bioassays, conducted with these nanomaterials at different concentrations, demonstrated concentration-dependent mortality and significant behavioral responses. The exposures with nanomaterials also resulted in a deposition of black soot on the larval cuticle. The most effective nanomaterial was found as ferrous oxide nanomaterial which induced 100% larval mortality as well as significant contact irritancy. The results indicate the potential use of GO and GG-based nanomaterials against Ae. aegypti after concentration optimization.

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## A Comprehensive Study to Unleash the Putative Inhibitors of Serotype2 of Dengue Virus: Insights from an In Silico Structure-Based Drug Discovery.

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Dengue fever is a mosquito-borne disease that claims the lives of millions of people around the world. A number of factors like disease's non-specific symptoms, increased viral mutation, growing antiviral drug resistance due to reduced susceptibility, unavailability of an effective vaccine for dengue, weak immunity against the virus, and many more are involved. Dengue belongs to the Flaviviridae family of viruses. The two species of the vector transmitting dengue are Aedes aegypti and Aedes albopictus, with the former one being dominant. Serotypes 2 of dengue fever are spread to the human body and cause severe illness. Recently, dengue has imposed an aggressive effect synergistically with the COVID-19 pandemic. As a result, we concentrated our efforts on finding a potential therapeutic. For this, we chose natural compounds to fight dengue fever, which is currently regarded as successful among many drug therapies.

Following this, we started the in silico experiment with 922 plant extracts as lead compounds to fight serotype 2. In this study, we used SwissADME for analyzing ligand drug-likeness, pkCSM for designing an ADMET profile, Autodock vina 4.2 and Swissdock tools for molecular docking, and finally Desmond for molecular dynamics simulation. Ultimately 45 were found effective against the 2'O methyltransferase protein of serotype 2. ChEMBL376820 was found as possible therapeutic candidates for inhibiting methyltransferase protein in this thorough analysis. Nevertheless, more in vitro and in vivo research are required to substantiate their potential therapeutic efficacy.

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## Dengue epidemiology and molecular characterization of vector mosquitoes; *Aedes aegypti* and *Aedes albopictus* in Mizoram, Northeast India.

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Dengue is the most important arboviral disease, and *Aedes aegypti* and *Aedes albopictus* are the major vectors. It is one of the most recent emerging diseases in Mizoram state with a fast-growing burden over the years. The present study aimed to explore the epidemiological patterns of dengue in Mizoram, and to identify vector mosquito species. In the present study, two years secondary data of dengue disease in Mizoram was studied. Dengue vector mosquitoes and dengue virus serotypes circulating in vector mosquitoes in Aizawl City were identified by sequence analysis of COX1 and C-prM genes respectively. In Mizoram State, the annual case prevalence of dengue was found to be 300.45 per 100,000 persons. Seasonality of the disease was also noted, with the peak season occurring from July to December. There is a significant positive relationship between age and the disease prevalence ( $p < 0.05$ ,  $r = 0.909$ ). DENV was detected in 10 pools of *Ae. aegypti* with a minimum infection rate (MIR) of 18.05, and 8 pools of *Ae. albopictus* with a minimum infection rate of 16.03. The viral isolates were found to correspond to serotypes 2 (DENV2) and 3 (DENV3). A total of 6 pools of *Aedes aegypti* and 4 pools of *Aedes albopictus* tested positive with DENV2 serotype, while 4 pools of *Aedes aegypti* and 3 pools of *Aedes albopictus* tested positive with DENV3 serotype. DENV2 serotype was more common, and multiple serotypes (DENV2 and DENV3) were also detected in two pools from one locality. There is an increasing trend in dengue infection in Mizoram, and seasonality of the disease was observed with the peak season during July to December. The higher incidence rate among the adults indicated that outdoor activities may be related to the high risk of the infection. However, lack of immunity among the older people may also be a possible reason of higher disease incidence rather than the activity pattern. Early identification and frequent monitoring of infected vector mosquitoes will yield an early warning indicator for forecasting dengue epidemics in the future.

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## Identification of *Aedes aegypti* and *Aedes albopictus* eggs based on image processing and elliptic fourier analysis.

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Dengue hemorrhagic fever is a worldwide epidemic caused by dengue virus and spread by infected female mosquitoes. The two main mosquito species vectors of the dengue virus are *Aedes aegypti* and *Aedes albopictus*. Conventionally, the identification of these two species' egg is time-consuming which makes vector control more difficult. However, although attempts on efficiency improvements by providing automatic identification have been conducted, the earliest stage is at the larval stage. In

addition, there are currently no studies on classifying to distinguish the two vectors during the egg stage based on their digital image. A total of 140 egg images of *Aedes aegypti* and *Aedes albopictus* were collected and validated by rearing them individually to become adult mosquitoes. Image processing and elliptic Fourier analysis were carried out to extract and describe the shape difference of the two vectors' eggs. Machine learning algorithms were then used to classify the shape signatures. Morphometrically, the two species' eggs were significantly different, which *Aedes albopictus* were smaller in size. Egg-shape contour reconstructions of principal components and Multivariate Analysis of Variance (MANOVA) revealed that there is a significant difference (p value [Formula: see text]) in shape between two species' eggs at the posterior end. Based on Wilk's lambda of the MANOVA results, the classification could be done using only the first 3 principal components. Classification of the test data yielded an accuracy of 85.00% and F1 score 84.21% with Linear Discriminant Analysis applying default hyperparameter. Alternatively, k-Nearest Neighbors with optimal hyperparameter yielded a higher classification result with 87.50% and 87.18% of accuracy and F1 score, respectively. These results demonstrate that the proposed method can be used to classify *Aedes aegypti* and *Aedes albopictus* eggs based on their digital image. This method provides a foundation for improving the identification and surveillance of the two vectors and decision making in developing vector control strategies.

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## Modeling the present and future distribution of arbovirus vectors *Aedes aegypti* and *Aedes albopictus* under climate change scenarios in Mainland China.

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*Aedes aegypti* and *Aedes albopictus* are two important mosquito species which transmit various infectious arbovirus diseases represented mainly by dengue fever. These two species of mosquito have a wide range of distribution and strong transfer capacity. With ongoing global climate change, we are facing an increasing public health threat from the rapid spread of vectors in wider geographical areas. Based on observed occurrence records of *Ae. aegypti* and *Ae. albopictus* and high-resolution environmental layers reflecting climate and land-use conditions, a Maxent niche modeling approach was adopted to model the current and future distribution of both species in Mainland China. Our models provide predictions of suitable habitat shifts under future climate scenarios up to the 2050s. Both species were predicted to expand their niche range to varying degrees under future climate scenarios. *Aedes aegypti* was modeled to expand its habitat from Guangdong, Guangxi, Yunnan and Hainan to Fujian, Jiangxi and Guizhou. *Aedes albopictus* was modeled to increase magnitude of distribution within its present range of northern, southwestern and southeastern coastal areas of Mainland China. Area and population exposed to mosquitoes are predicted to increase significantly. Environmental variables that have significant impact on the distribution of mosquitoes are also revealed by our model. The results of our study can be referenced in further ecological studies and will guide the development of strategies for the prevention and control of mosquito-borne diseases.

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## Scanning electron microscopy of damage caused by *Mesocyclops thermocyclopoides* (Copepoda: Cyclopoidea) on larvae of the Dengue fever vector *Aedes aegypti* (Diptera: Culicidae).

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Dengue fever is a mosquito-borne viral disease, whose main biological vector is *Aedes aegypti*. This mosquito colonizes tropical areas where the disease is endemic. The most obvious action against dengue is attacking its vector. Biological control appears to be an alternative approach, using natural enemies of the mosquitoes, such as predatory copepods. Thus, the morphological study of the damage caused by copepods is important to understand its predatory capacity. Twenty-five *A. aegypti* larvae were exposed to the copepod *Mesocyclops thermocyclopoides* and the damage caused by the copepods was evaluated using scanning electron microscopy. The larvae showed damage mainly at the anal segment, the siphon and the abdomen; only three attacks to the head were observed. The size of the siphon might be of importance in determining whether or not a copepod will attack a mosquito larva.

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### Vector competence of *Aedes albopictus* and *Aedes aegypti* (Diptera: Culicidae) for the DEN2-FJ10 and DEN2-FJ11 strains of the dengue 2 virus in Fujian, China.

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Dengue is an acute, emerging, infectious disease transmitted by *Aedes* mosquitoes that has become a serious global public health problem. The DEN2-FJ10 and DEN2-FJ11 strains of the dengue 2 virus were originally isolated from the serum of a patient with dengue fever in Fujian Province, China, in 1999. Our data provide the first assessment of the vector competence of *Aedes* mosquitoes with respect to the DEN2-FJ10 and DEN2-FJ11 strains of the dengue virus. There were significant differences in the replication rates of these two viral strains in *Aedes albopictus* and *Aedes aegypti* ( $P < 0.05$ ). In summary, our results indicate that *Ae. albopictus* and *Ae. aegypti* mosquitoes are moderately competent vectors of the DEN2-FJ10 and DEN2-FJ11 strains of the dengue virus and provide the first evidence of the effect of these two viral strains on the vector competence of mosquitoes in China.

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